

The Logarithmic Ceiling: Dose Escalation and the Limits of Durability in Intravitreal Anti-VEGF Therapy: A Narrative Review

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Abstract

Aflibercept 8 mg delivers a fourfold higher molar dose than the 2 mg formulation and has been evaluated in neovascular age-related macular degeneration (nAMD), diabetic macular oedema (DME), polypoidal choroidal vasculopathy (PCV), and, most recently, macular oedema secondary to retinal vein occlusion. Interval extension is broadly similar across the three indications for which mature ninety-six-week data exist, despite substantial differences in the degree to which their exudation is VEGF-dependent. This review argues that such similarity is the expected result rather than a coincidence, and that its implications for drug development are more important than its implications for prescribing. Under first-order elimination at fixed clearance, the additional time a drug spends above any suppression threshold following a dose increase of factor F is $t_{1/2} \times \log_2(F)$. The threshold cancels, so the increment purchased is a property of the molecule and the dose ratio, not of the disease. Two consequences follow. First, comparable durability across pathophysiologically distinct conditions requires no disease-specific explanation. Second, the relationship is logarithmic: the fourfold increase from 2 mg to 8 mg purchases approximately two elimination half-lives, a further doubling to 16 mg would purchase one, and doubling the eight-week interval by molar dose alone would require roughly thirty-fold the standard dose. Observed extension substantially exceeds this prediction, and the leading explanation, derived from pharmacokinetic modelling rather than from direct measurement, attributes much of the excess to reduced ocular clearance of the concentrated formulation rather than to the molar increase itself. On that decomposition the molar step accounts for between a third and a half of the durability actually achieved. That explanation does not rescue dose escalation. It relocates the operative variable from how much drug is given to how long it persists. Both readings converge on the same conclusion: molar escalation is an exhausted lever, and further durability must come from altering half-life through delivery or from altering the threshold through mechanism.

Keywords: aflibercept 8 mg; intravitreal pharmacokinetics; treatment durability; dose escalation; anti-VEGF therapy

1. Introduction

Anti-VEGF therapy for exudative retinal disease is limited less by efficacy than by frequency. Injection burden drives attrition, undertreatment, and the persistent gap between trial and real-world visual outcomes. Every major development programme of the past decade has therefore targeted durability rather than potency, and higher-dose formulations represent the most direct expression of that strategy.

Aflibercept 8 mg administers 0.07 mL at 114.3 mg/mL against 0.05 mL at 40 mg/mL for the 2 mg formulation: a fourfold higher molar dose of an unmodified molecule with unchanged binding characteristics [1]. Its evaluation now spans four diseases. CANDELA established anatomic signal in nAMD [2], PULSAR demonstrated non-inferiority at 12- and 16-week intervals in nAMD [3,4], PHOTON did so in DME [5,6], a post hoc analysis addressed PCV [7], and QUASAR supported approval for macular oedema secondary to retinal vein occlusion in late 2025 [8].

This breadth permits an inference not usually available. The intervention is pharmacologically identical across all four conditions; the diseases are not. Comparing durability across them therefore isolates what dose escalation contributes independently of pathophysiology. The finding is that it contributes similarly everywhere, and the argument developed here is that this similarity, correctly interpreted, identifies the point at which the dose-escalation strategy stops paying.

This is a narrative rather than a systematic review. The evidence base was assembled by searching PubMed and Google Scholar to August 2026 for intravitreal aflibercept 8 mg across all indications, supplemented by the pivotal trial reports and their published post hoc analyses, the current United States prescribing information, and the reference lists of retrieved articles. Priority was given to randomised evidence and to real-world series reporting interval outcomes with defined denominators.

2. Durability across indications

Cross-trial comparison requires care, because the pivotal trials define durability differently. PHOTON reports the proportion of completers who did not require shortening below a threshold interval. PULSAR reports the last assigned dosing interval and the proportion qualifying for further extension. These constructs are related but not interchangeable, and much of the apparent variation across indications dissolves once the distinction is held in view.

Table 1. Aflibercept 8 mg durability outcomes at week 96.

	nAMD (PULSAR) [4]	DME (PHOTON) [6]
<i>As reported in the primary publications</i>		
≥12-week interval	87% of 8q12 arm (last assigned)	89% maintained
≥16-week interval	78% of 8q16 arm qualified	83% maintained
≥20-week interval	53% of 8q16 arm qualified	44% qualified
24-week interval	31% of 8q16 arm	27% eligible
BCVA change	+5.6 (8q12), +5.5 (8q16), +6.6 (2q8) letters	Non-inferior to 2q8
Injections to week 96	9.7 (8q12), 8.2 (8q16) vs 12.8 (2q8)	Approximately 6 fewer than 2q8
Intraocular inflammation	Comparable to 2 mg	1.2% in both arms
<i>Harmonised to week-96 completers, last assigned interval (label-derived) [1]</i>		
≥16-week interval	71%	72%
≥20-week interval	47%	44%

Across most thresholds the two indications lie within a few percentage points of one another, and the residual differences run in inconsistent directions: DME is marginally higher at 12 and 16 weeks, nAMD marginally higher at 20 and 24 weeks. The widest gap, nine points at the 20-week threshold, narrows to three when both datasets are harmonised to a common completer denominator (Table 1). Because the two trials define the endpoint differently, this supports the weaker claim that no large difference is demonstrable rather than the stronger claim that durability is identical. Design asymmetries plausibly account for what remains. PHOTON prohibited extension during the first year and randomised 1:2:1 in favour of the 8q12 arm, whereas PULSAR permitted earlier extension and randomised 1:1:1. Baseline populations differed in age and disease chronicity.

The lower block of Table 1 reports both datasets harmonised to a single completer-based definition, as the current prescribing information does. These label-derived figures are not reported in this form in either primary publication and are given for comparability rather than as independent evidence; they are quoted here because no peer-reviewed source presents the two trials on a common denominator.

PCV is consistent with the same pattern, with the caveat that the evidence is weaker. In the PULSAR post hoc analysis, visual acuity improved by 9.5 and 8.4 letters in the 8q12 and 8q16 arms against 9.1 letters for 2 mg every eight weeks, and polypoidal lesions were absent in 37%, 47%, and 38% of completers respectively [7]. For context, complete polyp regression with aflibercept 2 mg monotherapy was 39% at week 52 in PLANET [9], and a small real-world series reported 62.5% with aflibercept 8 mg against 75.0% with brolucizumab 6 mg after three

monthly doses [10]. Dose escalation therefore does not appear to increase lesion-level regression. The PULSAR analysis is a subgroup of a trial not powered for it, and pooled real-world synthesis in this subtype rests on few studies with heterogeneous regression definitions [11]. It is corroborative rather than confirmatory.

Three diseases with materially different pathophysiology, then, receive materially similar interval extension from the same dose ratio.

3. Why the increment is disease-independent

Aflibercept binds VEGF-A, VEGF-B, and placental growth factor as a soluble decoy receptor. Exudation is suppressed while free intravitreal drug remains above the concentration required to neutralise local VEGF production, a quantity here termed the suppression threshold. Two distinct properties of the concentration-time curve govern two distinct clinical outcomes, and their conflation is the source of the puzzle.

Depth is the peak concentration relative to threshold, and it determines whether an eye dries at all. Quadrupling the molar dose raises the peak fourfold, so eyes in which a standard dose cleared the threshold only marginally are carried well beyond it, and exudation that persisted under 2 mg resolves. Depth scales linearly with dose, manifests within the first cycle, and is disease-sensitive, because lesion biology sets the threshold.

Duration is the time spent above threshold, and it determines the achievable interval. Under first-order elimination, concentration declines exponentially, so time above a fixed threshold is $t = (t_{1/2} / \ln 2) \times \ln(C_0/C_{\min})$. Multiplying the dose by factor F multiplies C_0 by F , and the additional time above the same threshold is:

$$\Delta t = t_{1/2} \times \log_2(F)$$

The threshold term cancels. The increment depends only on the elimination half-life and the dose ratio, neither of which varies by disease. A fourfold increase buys two half-lives whether the eye contains a mature fibrovascular membrane, a diabetic capillary bed, or a polypoidal lesion. This holds so long as the elimination half-life is itself unchanged by the dose increase, an assumption the aflibercept 8 mg data do not straightforwardly satisfy and which Section 4.1 examines.

Measured human ocular half-life of free aflibercept is approximately 11 days [12], derived from a single small pharmacokinetic study, which places two half-lives at roughly three weeks. Supporting measurements are consistent in direction: aqueous VEGF-A remains suppressed below assay limits for approximately ten weeks after 2 mg [13], and aflibercept's suppression duration approximately doubles that of ranibizumab [14]. Mechanistic modelling of VEGF suppression built on these parameters predicts that extended fixed intervals remain feasible with the higher dose [15,16].

The central claim requires only the ratio, not the absolute value. Even if the half-life estimate is revised, the increment remains a function of drug and dose rather than of disease, and comparable durability across indications remains the predicted result.

4. The logarithmic ceiling

The same equation that explains parity also constrains the future of the strategy, and this is its more consequential implication. Because the relationship is logarithmic, each successive doubling of dose purchases one additional half-life, irrespective of where the doubling starts. The step from 2 mg to 8 mg, a fourfold increase, purchased two half-lives. A further doubling to 16 mg would purchase one, approximately eleven days, for twice the protein load in an eye already receiving a concentrated formulation. Table 2 sets out the full schedule; it holds the elimination half-life constant across doses, an assumption examined in Section 4.1.

Table 2. Interval extension purchased by molar dose escalation, assuming first-order elimination at fixed clearance and $t_{1/2} = 11$ days. Observed values describe the fixed intervals assigned in the pivotal trials, not titration to failure.

Dose relative to 2 mg	Half-lives gained	Predicted additional interval	Observed
2× (4 mg)	1	~1.6 weeks	Not studied
4× (8 mg)	2	~3.1 weeks	+4 to +8 weeks in treatment-naïve trials
8× (16 mg)	3	~4.7 weeks	Not studied
16× (32 mg)	4	~6.3 weeks	Not studied
34× (~68 mg)	5.1	~8 weeks	Not studied

The final row is the informative one. Doubling a standard eight-week interval to sixteen weeks by molar dose alone would require approximately thirty-fold the 2 mg dose. No intravitreal formulation of a 115 kDa fusion protein can deliver that in a volume the eye tolerates, and the inflammatory consequences of attempting it are readily foreseeable given that a fourfold increase has already generated a real-world inflammation signal (Section 5). Dose escalation is not merely subject to diminishing returns; it is subject to diminishing returns steep enough that further molar increase has little left to offer.

4.1 The observed excess, and what it implies for Table 2

Observed extension exceeds the prediction. Moving from eight-week to twelve- or sixteen-week intervals represents four to eight additional weeks against a predicted three. One qualification applies immediately. PULSAR and PHOTON assigned fixed twelve- and sixteen-week regimens rather than titrating each eye to failure, so these figures record what the protocols assigned and patients tolerated rather than the maximum the molecule can sustain; the true achievable interval may be longer, in which case the excess is larger than stated. The discrepancy nonetheless requires accounting, and the accounting matters more than the discrepancy.

The leading explanation is that ocular clearance of the 8 mg formulation is lower than that of the 2 mg formulation by approximately one third, corresponding to six to nearly nine additional weeks above ocular VEGF targets [17], a finding subsequently discussed in commentary [18]. It is a modelled rather than a measured ocular clearance, derives from a conference abstract that has not appeared in peer-reviewed full publication, and is attributed to formulation properties such as viscosity and altered intravitreal distribution rather than to the molar increase itself. It should be treated as provisional.

If that estimate is correct it has a direct consequence for Table 2 which must be stated plainly. Half-life varies inversely with clearance, so a reduction in clearance of one third implies a half-life roughly fifty per cent longer for the concentrated formulation. Table 2 holds half-life constant and therefore describes the yield of molar escalation only at fixed formulation properties. It is not a universal ceiling on durability; it is a ceiling on what additional molar dose can contribute when nothing else changes.

This does not weaken the argument. It sharpens it, provided the accounting is done by subtraction rather than by addition. The modelled six to 8.9 weeks is the total difference in time above target between two formulations that differ in both molar dose and clearance, so the molar contribution sits inside that total rather than alongside it. Subtracting the approximately 3.1 weeks attributable to the fourfold increase leaves roughly three to six weeks attributable to clearance. On that decomposition the molar increase accounts for between a third and a half of the durability gain, and the remainder is a formulation property that further molar escalation would not by itself reproduce. The lever that worked is only partly the one the dose number describes.

A second qualification points the same way. Aflibercept elimination is not strictly first-order. It proceeds through target-mediated disposition via binding to endogenous VEGF as well as through proteolysis, a saturable process, and systemic data show dose-dependent clearance that approaches linearity only at higher concentrations. The $\log_2$ relation is therefore a first approximation rather than a mechanism. Where target-mediated elimination is saturable, the benefit of adding drug beyond saturation is inherently self-limiting.

Under either qualification the conclusion holds, and is reached by a shorter route. Adding molar dose to an already-saturating concentration, at fixed formulation, is the least promising available direction.

5. Safety and the cost of the fourfold step

Trial safety has been reassuring and consistent. Intraocular inflammation occurred in 1.2% of both 8 mg and 2 mg arms in PHOTON [6], and a 96-week pooled analysis across PULSAR and PHOTON found no long-term intraocular pressure or glaucoma signal relative to standard dose [19].

Real-world data are less uniform, and the divergence is material to any argument about escalating further. A systematic review of 16 studies comprising 2,019 eyes and 9,481 injections found low inflammation rates in trials but higher and more variable rates in practice [20]. Case series have reported non-infectious inflammation with retinal vasculitis in three of 35 injections [21], and in nine of 83 eyes during the loading phase, equivalent to 10.8% of eyes or 4.04% of 223 injections [22], with further multicentre series describing similar clusters [23,24]. A large real-world cohort of 8,675 injections in 1,429 eyes reported low event rates [25], and pharmacovigilance disproportionality analyses show elevated reporting odds for inflammatory events with the concentrated formulation [26], though such analyses reflect reporting behaviour and cannot establish incidence.

The relevant inference is not that aflibercept 8 mg is unsafe; the weight of evidence indicates it is not. It is that a fourfold increase in intravitreal protein load has produced a detectable and reproducible inflammatory signal in real-world use, while the molar increase itself accounts for only about three weeks of the durability gained. The ratio of tolerability cost to durability benefit worsens with each further doubling, because the benefit is logarithmic while the protein load is linear.

No dataset stratifies inflammation risk by indication. Because real-world signals derive predominantly from nAMD cohorts while the largest trial denominator sits in DME, any apparent difference is confounded by setting, and the question should be recorded as unresolved.

6. What the model predicts, and where it should be tested

A framework that explains existing observations earns confidence only by predicting departures. Two predictions are set out below, together with the analysis that would test both.

Elevated threshold should attenuate the increment. The threshold is not fixed within a disease. Chronic lesions are larger, more mature, pericyte-invested, and accompanied by macrophage-driven remodelling and fibrosis, with a component of exudation that is VEGF-independent and therefore inaccessible to any dose [27]. The same increment applied above a raised threshold, in eyes with a shorter starting interval, should yield less.

Pooled real-world data from previously treated eyes conform. Interval extension is +1.79 weeks (95% CI +1.32 to +2.27), below the predicted increment and well below naive-trial performance, while anatomic response persists: central subfield thickness falls by 21.5 μm , pigment epithelial detachment height by 22.9 μm , intraretinal and subretinal fluid resolve in 37.5% of eyes each, and visual acuity remains stable at +0.83 letters [28]. Individual switch cohorts show the same pattern [29,30,31]. Depth is preserved because a fourfold peak still drives many eyes below threshold; duration is attenuated because the increment is applied above a raised threshold.

Two design factors depress the observed figure further and are rarely reported in a form permitting adjustment. Treat-and-extend protocols with two-week increments require completion of a shorter interval before a longer one can be documented, so an eye beginning at six weeks

cannot appear on a sixteen-week interval before approximately week 50, whereas most switch cohorts were followed for less. Separately, label-mandated minimum intervals convert eyes requiring more frequent dosing into discontinuations rather than dose reductions, and all-cause discontinuation in switch cohorts approaches 20% [28]. The first biases observation downward, the second upward among survivors.

Continuously regenerated VEGF should compress the ceiling. Where ischaemia sustains VEGF production throughout the dosing cycle, additional depth should buy proportionally less duration. Macular oedema secondary to retinal vein occlusion is the natural test, and the current labelling permits eight-week dosing after loading, against intervals up to twenty weeks in nAMD and DME [1]. This observation is suggestive but not probative: QUASAR compared eight-week dosing against four-week standard dosing and did not test whether longer intervals were attainable [8], so the labelled interval reflects the question asked as much as the biology. A trial permitting extension in retinal vein occlusion would resolve it.

Testing either prediction properly requires eye-level rather than cohort-level analysis, and specifically requires determining whether anatomic response and interval achievement are coupled within individual eyes. The framework predicts weak coupling. No published dataset has examined it, and any adequately followed cohort could.

7. Implications

For clinical practice, the expectation should be a bounded and roughly constant increment. Aflibercept 8 mg reliably buys interval extension across exudative disease, and the magnitude in a given eye depends less on which disease is being treated than on how far above its suppression threshold that eye has been driven. Two to three fewer injections per year in a previously treated eye is the realistic and worthwhile outcome, and framing it against naive-trial intervals misrepresents both.

For trial reporting, the field should converge on a single durability definition. The difficulty in constructing Table 1 was not scarcity of data but incommensurability of endpoints, and the same problem afflicts anatomic outcomes, where prespecified clinical moderators failed to reduce residual heterogeneity below approximately 55% in pooled analysis [28]. Reporting achieved interval with an explicit denominator and time origin, as time to first achievement rather than a cross-sectional proportion, would make cross-trial synthesis possible. Prospective real-world programmes now underway across both major indications offer an opportunity to standardise this prospectively [32].

For drug development, the implication is the substantive one. Durability under fixed-ratio dose escalation is logarithmic in dose and independent of disease, and the observed performance of aflibercept 8 mg appears to owe more to reduced clearance than to the molar increase. The remaining levers are therefore structural rather than quantitative: extending half-life through sustained-release delivery, reducing clearance through formulation, or lowering the threshold by

addressing the VEGF-independent component of exudation that no dose reaches. Programmes pursuing the first two are already the ones producing meaningful interval gains. Programmes contemplating further molar escalation are, on this analysis, pursuing one additional half-life per doubling at linearly increasing immunological cost.

8. Limitations

Several constraints bear on this argument. The human ocular half-life used throughout derives from a single small pharmacokinetic study [12], and the framework's absolute predictions scale directly with it, although its central claim of disease-independence does not. The clearance estimate on which Section 4.1 turns comes from a conference abstract [17] and has been neither independently measured nor peer-reviewed in full. Aflibercept elimination is target-mediated and therefore not strictly first-order, so the $\log_2$ relation is an approximation whose accuracy at high concentration is unknown. All week-96 durability figures are complete analyses, and the two pivotal trials define the endpoint differently, so cross-indication comparison rests on partially incommensurable quantities; the harmonised figures that address this are label-derived rather than independently published. The pivotal trials assigned fixed dosing intervals rather than titrating to failure, so the durability figures throughout describe assigned and tolerated intervals rather than maximal attainable ones. Evidence in PCV is confined to a post hoc subgroup and small real-world series, and evidence in retinal vein occlusion is limited to a single trial that did not test extension. Finally, the framework's central prediction, that anatomic response and interval achievement are weakly coupled within individual eyes, has not been tested in any published dataset.

9. Conclusion

The near-equivalence of aflibercept 8 mg durability across neovascular age-related macular degeneration, diabetic macular oedema, and polypoidal choroidal vasculopathy is not evidence that these diseases share a mechanism. It is evidence that interval extension under dose escalation is pharmacokinetic rather than pathophysiologic: the dose ratio and elimination profile set the increment, while disease biology sets the threshold and thereby determines how many eyes benefit and how deeply, but not by how much the interval moves.

The consequence is a ceiling rather than a plateau. At fixed formulation, each doubling of molar dose purchases one elimination half-life, so the fourfold step already taken delivered most of what molar escalation can deliver, and doubling a standard interval by dose alone would require a formulation the eye cannot accept. If the observed durability advantage of the concentrated formulation reflects reduced clearance rather than molar increase, as the available modelling suggests, then the lesson is sharper still: the productive variable was never how much drug is delivered but how long it stays. That is a problem for delivery science and for target selection, and it is where the next durable gains will be found.

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